Appl. Phys. B 74 [Suppl.], S259–S263 (2002)
DOI: 10.1007/s00340-002-0882-z

Applied Physics B
Lasers and Optics

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High-precision measurement of angular dispersion in a CPA laser

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Received: 15 September 2001/
Revised version: 20 December 2001
Published online: 27 June 2002 •

ABSTRACT Angular dispersion is measured with the accuracy of 0.2 μrad/nm by a new method based on spectrally resolved interferometry. It is linear, simple and allows for real-time alignment of the stretcher-compressor system of CPA lasers.

PACS 42.60.v; 42.65.Re; 95.75.Kk

1 Introduction

In a complex optical system, such as a short pulse chirped pulse amplification (CPA) laser [1], there are various causes of angular dispersion. The prism pair in the oscillator may have a slightly different apex angle or, more significantly, non-parallel refractive surfaces. Slightly wedged optical components, like the output coupler, the laser crystals, or glass filters, also cause angular dispersion, not to mention the non-parallelism of the gratings in the stretcher or compressor [2, 3].

One of the most crucial alignment procedures of CPA lasers is the accurate setting of the stretcher-compressor system [2–5]. Although the effect of non-parallelism of the gratings on the phase modulation of the compressed pulse can be partially compensated for by changing the grating separation, the residual angular dispersion results in a tilted pulse front [6–9], and hence leads to an increased pulse duration and reduced intensity when the beam is focused onto a target [10, 11]. Since the effect of residual angular dispersion becomes increasingly important at shorter pulse duration, sub-20 fs laser systems call for more accurate and in situ diagnostics than existing systems can provide [2, 11–13]. Previous techniques such as using a spatially reversed Mach–Zender [11] and Michelson [13] interferometer, or a tilted pulse-front autocorrelator [12], allow for measurement with the accuracy of several μrad/nm. A further disadvantage of the first two methods is that they require multiple-shot measurements. The method of Sacks et al. [12] is suitable for simultaneous monitoring of pulse length and angular dispersion, but requires high intensities for the second harmonic generation, and therefore it is not sensitive enough for measuring the stretched pulse.

In this paper, we propose a new single-shot method based on spectrally resolved interference (SRI) [14–16] to measure the residual angular dispersion of a laser beam. The technique is very simple, linear and allows for high-precision real-time monitoring of angular dispersion.

2 Theory

Let us assume two monochromatic waves

$$\begin{aligned}\underline{E}_1 &= E_1 \cos\left[\underline{k}_1\,\underline{r} + \varphi_1\right]\\ \underline{E}_2 &= E_2 \cos\left[\underline{k}_2\,\underline{r} + \varphi_2\right]\end{aligned} \tag{1}$$

travelling in the directions $\underline{k}_1$, $\underline{k}_2$ at a small angle $\varepsilon = \varepsilon_2 - \varepsilon_1$ to each other, and interfering on a vertical plane that is observed along the y-axis (see Fig. 1). The interference is constructive, when the interference term

$$\left\langle \underline{E}_1\, \underline{E}_2 \right\rangle \propto \cos\left[\left(\underline{k}_2 - \underline{k}_1\right)\underline{r} + \Delta\varphi\right] = \cos\left[\frac{2\pi}{\lambda}\varepsilon\, y + \Delta\varphi\right] \tag{2}$$

has a maximum, where $\Delta\varphi = \varphi_2 - \varphi_1$. In (2) small-angle approximation has been used. We define the distance between two consecutive bright fringes as the line-separation Λ. This definition is equivalent to

$$\cos\left[\frac{2\pi}{\lambda}\varepsilon\,(y+\Lambda) + \Delta\varphi\right] = \cos\left[\frac{2\pi}{\lambda}\varepsilon\, y + \Delta\varphi\right], \tag{3}$$

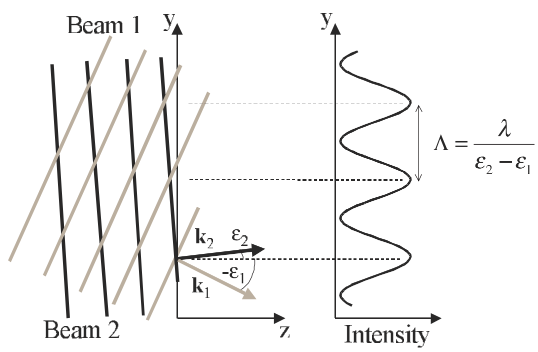

FIGURE 1 Interference of two monochromatic waves travelling at angles to the z-axis. The line separation Λ is determined by the angle of beams

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thus the angle of the two beams and the line separation is related by

$$\varepsilon = \frac{\lambda}{\Lambda} \,. \tag{4}$$

For a broadband beam the superposition of the interference patterns belonging to each spectral component leaves a blurred image with poor fringe visibility [15]. However, with the help of spectral resolution of the pattern, the two interfering beams can be regarded as a collection of independent monochromatic waves "placed" side-by-side. As a result of the interference of two broadband waves propagating at an angle ε, the position of the nth intensity maximum is then given by

$$y(\lambda) = \left(n - \frac{\Delta\varphi(\lambda)}{2\pi}\right)\frac{\lambda}{\varepsilon} \,. \tag{5}$$

When no angular dispersion is present in the beam, this equation describes straight lines splayed out towards the longer wavelength side of the spectrum. The line-separation is given as

$$\Lambda(\lambda) = |y_n(\lambda) - y_{n-1}(\lambda)| = \frac{\lambda}{|\varepsilon|} \,. \tag{6}$$

Thus, spectral resolution of the interference pattern allows for determining the angle of two broadband beams.

To incorporate angular dispersion into the model, we should consider a more general case. Two beams travelling at arbitrary angles, measured from the z-axis counter-clockwise, interfere in the x–y plane. When angular dispersion is present in a beam, its spectral components travel in different directions, i.e. $\varepsilon = \varepsilon(\lambda)$. As the wavelength-dependence is usually small, we shall consider the Taylor expansion around the central wavelength λ_0:

$$\begin{aligned}\varepsilon(\lambda) &= \varepsilon_2(\lambda) - \varepsilon_1(\lambda)\\ &= \varepsilon_2(\lambda_0) - \varepsilon_1(\lambda_0) + \left(\varepsilon_2'(\lambda_0) - \varepsilon_1'(\lambda_0)\right)(\lambda - \lambda_0)\\ &\quad + \frac{\left(\varepsilon_2''(\lambda_0) - \varepsilon_1''(\lambda_0)\right)}{2}(\lambda - \lambda_0)^2 + \ldots ,\end{aligned} \tag{7}$$

where $\varepsilon_1(\lambda_0)$ and $\varepsilon_2(\lambda_0)$ are the signed angles of the central wavelength component of each beam with the z-axis, $\varepsilon_1'(\lambda_0)$, $\varepsilon_2'(\lambda_0)$ are the first, $\varepsilon_1''(\lambda_0)$, $\varepsilon_2''(\lambda_0)$ are the second derivatives with respect to wavelength at the central wavelength. Equation (6) is then replaced by

$$\frac{\lambda}{\Lambda(\lambda)} = \left|\varepsilon(\lambda_0) + \varepsilon'(\lambda_0)(\lambda - \lambda_0) + \frac{\varepsilon''(\lambda_0)}{2}(\lambda - \lambda_0)^2 + \ldots\right| , \tag{8}$$

where $\varepsilon(\lambda) = \varepsilon_2(\lambda) – \varepsilon_1(\lambda)$.

The effect of angular dispersion on spectrally resolved interference fringes is demonstrated via numerical simulations along (2) with (8). The interference of two beams was calculated in the spectral range 750–850 nm. As (8) determines only the sum of the dispersions in the two beams, for simplicity we have restricted angular dispersion into beam 1 ($\varepsilon_1' \neq 0$, $\varepsilon_2' = 0$ and $\varepsilon_1'' = \varepsilon_2'' = 0$).

For the case of no angular dispersion (Fig. 2a), the lines in the pattern are virtually parallel, as expected from (5). When angular dispersion is present in the beam, the interference fringes are curved and become more distinctively splayed (Fig. 2b). With angular dispersion of the opposite sign, lines are splayed out towards the other end of the wavelength range (Fig. 2c).

We should note here that chirp, represented by $\Delta\varphi(\lambda)$ in (5), also causes a distortion of the spectrally resolved interference pattern (see [14, 16] for details), but it has no effect on line-separation. In the simulation above, all beams had zero chirp. On the other hand, it is important to see that the line spacing is unaffected by the dispersion of the beam (see (6)).

Figure 2d shows the $\lambda/\Lambda(\lambda)$ functions corresponding to the simulated SRI patterns. The plots are linear, as the beams possess no higher-order angular dispersion. The slope of the lines gives the value of angular dispersion.

We must add here that the sign of angular dispersion depends upon the choice of beam 1 and 2. With the interchange of beams, angular dispersion changes sign, as expected from (7). For determination of the sign of angular dispersion, see Sect. 4.

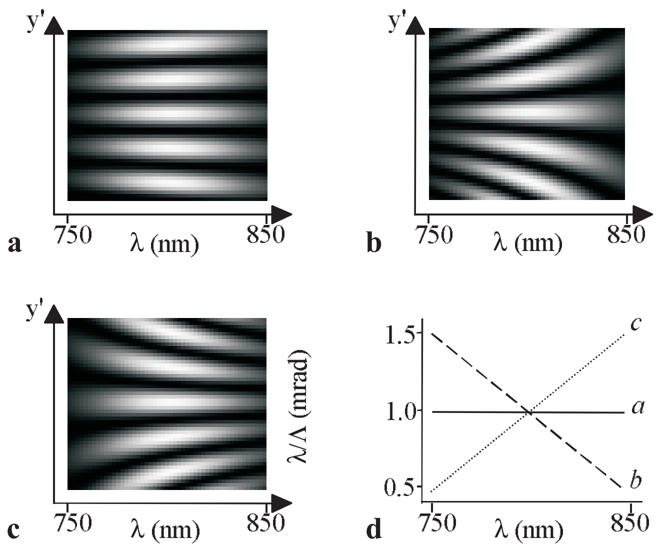

FIGURE 2 Simulated SRI fringes for pulses **a** without angular dispersion of **b** $\varepsilon_1' = 10.0$ μrad/nm and **c** $\varepsilon_1' = -10.0$ μrad/nm, and **d** the corresponding $\lambda/\Lambda(\lambda)$ plot

3 Experimental

For the measurements, the spectrally resolved interferograms from a Mach–Zender interferometer were used (see Fig. 3). Angular dispersion has been introduced into the 800-nm beam of our 72-MHz repetition rate Ti:S oscillator by a 45° fused-silica prism. The 20-fs pulses with a 55-nm bandwidth (FWHM) enter the interferometer through a beamsplitter plate, then in one arm fall onto an aluminium mirror which ensures constant reflectivity through the spectral range at a non-specific angle. In the other arm, by means of two mirrors at 45° to each other, the left and right sides of the beam are

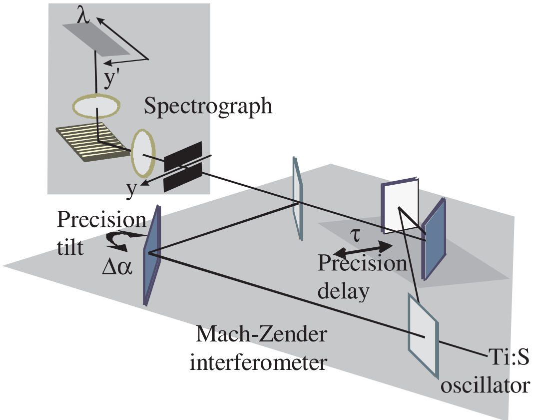

FIGURE 3 Experimental setup

interchanged, therefore we measure the angular dispersion of the beam twice, since

$$\varepsilon_1(\lambda) = \varepsilon_1(\lambda_0) + \varepsilon_1'(\lambda_0)(\lambda-\lambda_0) + \frac{\varepsilon_1''(\lambda_0)}{2}(\lambda-\lambda_0)^2 + \ldots$$
$$\varepsilon_2(\lambda) = \varepsilon_2(\lambda_0) - \varepsilon_1'(\lambda_0)(\lambda-\lambda_0) - \frac{\varepsilon_1''(\lambda_0)}{2}(\lambda-\lambda_0)^2 + \ldots. \quad (9)$$

This arrangement also ensures that, while the mirrors are translated along the bisector, when changing delay-time, the output beam leaves in the same direction. The interferometer should be set with equal arm lengths with the precision of 100 fs to see any interference at all [15]; otherwise, the change in delay has no effect on fringe spacing, therefore no effect on the result. We emphasize that this setup also eliminates the effect of chirp on interference patterns, as the two interfering beams have no relative chirp to each other; therefore, the setup will work with strongly chirped pulses.

The interference image is resolved by a home-made grating spectrograph (600 mm^{-1}), with the entrance slit parallel to the table, i.e. the plane in which the two sides of the beam has been interchanged. The spectrally resolved interference pattern is recorded by a CCD camera (EDC1000-HR Electrim Corp., 244 × 753 pixels). The low dynamic range of the CCD chip (8 bits) and the 10-ms integration time slightly smoothes the recorded pictures.

Spectrally resolved interference patterns of the same incoming beam have been recorded at different settings of the interferometer (see Fig. 4). These settings are: $\varepsilon_0 = 0.84$ mrad for patterns a and b and $\varepsilon_0 = -1.02$ mrad for c and d. The relative delay between left and right pictures is 30 fs.

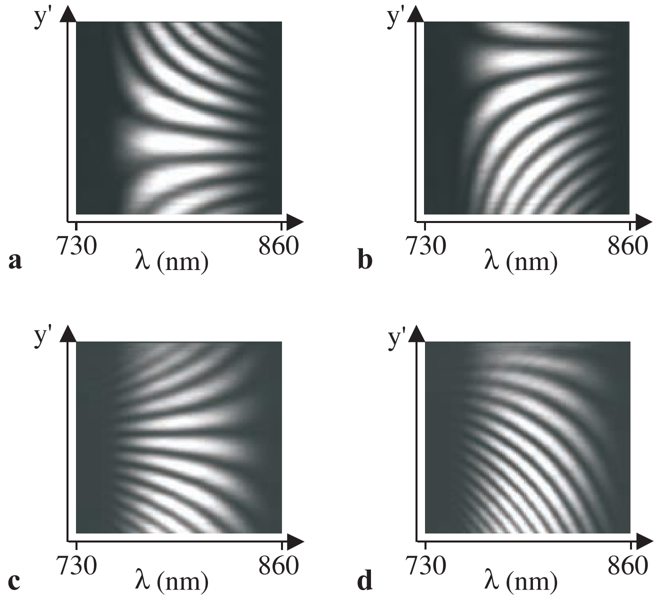

FIGURE 4 SRI patterns recorded at different delay times ($\Delta\tau = 30$ fs) and tilt angles ($\varepsilon_{a,b} = 0.81$ mrad, $\varepsilon_{c,d} = -1.02$ mrad) of the interferometer mirrors. For the measurement results, see Table 1

One may notice the following: in the top pictures the fringes are splayed towards the shorter wavelengths, and in the bottom pictures towards the longer wavelengths; as expected, the corresponding tilt angles have opposite signs. The fringes are more frequent in the bottom pictures, as the magnitude of the tilt angle is larger. On changing the delay time between the two arms of the interferometer, the pattern shifts vertically, as was indicated in [15].

4 Evaluation

4.1 *Coarse method*

As mentioned above, the line separation is inversely proportional to the angle of the two beams, so it becomes infinite for parallel phase fronts. When angular dispersion is present in a beam, it may be that phase fronts are parallel only in a certain wavelength. In this case, the spectrally resolved interference fringes become hyperbola-like (see Fig. 5), and with a calibrated spectrograph the wavelength of parallel phase fronts can be easily determined. On

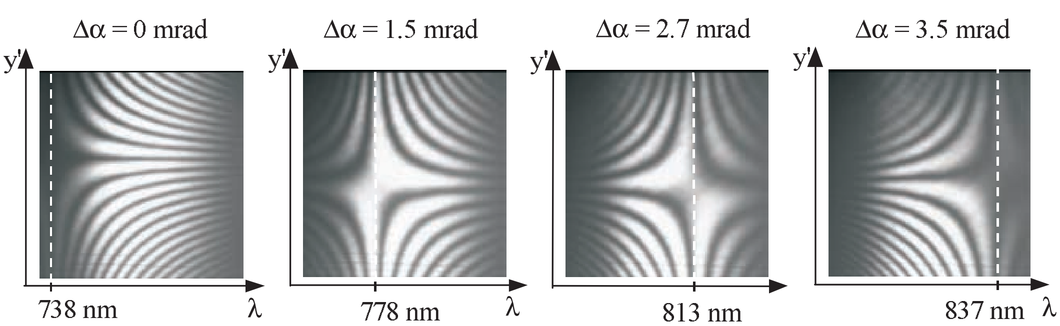

FIGURE 5 SRI pattern when phase fronts are parallel for certain wavelengths (indicated by *dotted lines*)

tilting the reference mirror by angle $\Delta\alpha$ around an axis perpendicular to y, the interferogram shifts as different wavelengths satisfy the parallel phase front criteria.

We observe that the pattern originally splayed towards the short-wavelength side changes into being splayed towards the long-wavelength side. This is caused by the sign change of the relative angle of the two beams. In Fig. 6 the relative tilt angle $\Delta\alpha$ of the reference mirror is plotted against the wavelength. As before, the slope of the function directly gives the value of first-order angular dispersion. We note that this procedure has a limited precision ($\sim 5\ \mu$rad/nm), but yields a quick qualitative evaluation. The method is effective for angular dispersion larger than 10 μrad/nm, up to 300 μrad/nm. The lower limit is defined by the precision of the tilting mount: while the pattern shifts along the spectral range, the tilt of the mirror must be determinable. The measurement also has an upper limit: for a large angular dispersion the hyperbolas of the pattern become too frequent to be resolved.

The sign of the angular dispersion is determined as follows: it is positive if the hyperbola-like pattern shifts toward the longer wavelength range when the mirror is tilted counterclockwise.

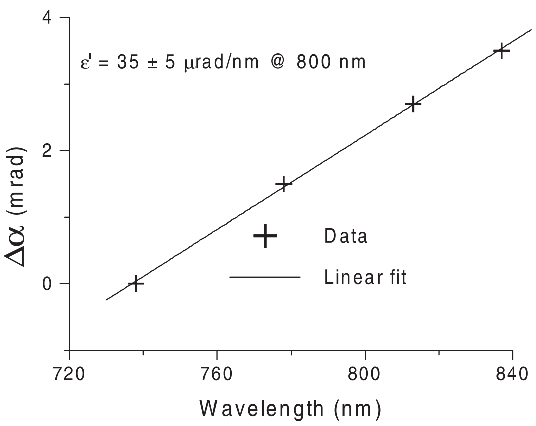

FIGURE 6 Relative tilt angle of reference mirror and the corresponding wavelength of parallel phasefronts

4.2 *Fine method*

With the aid of data processing using the digital image of the CCD camera, a more precise evaluation is possible. Each vertical section of the interference image is considered to be the result of the interference of two monochromatic beams, producing an intensity function of $\cos\left(2\pi y'\big/\Lambda(\lambda)\right)$ (see (2) and (6)), where y' is the conjugate coordinate to y. Thus, after normalization of an interference pattern, a cosine function is fitted to each vertical section of the image to obtain the line spacing function: $\Lambda = \Lambda(\lambda)$ (see Fig. 7). Then we plot the $\lambda\big/(2\Lambda(\lambda))$ function that gives the wavelength-dependent angle of the beams, providing information on angular dispersion. We found, that the relatively wide spectrum required quadratic approximation, therefore first- and second-order angular dispersion (ε', ε'') has been determined, i.e. in (7) and onwards, second-order Taylor expansion is used. Figure 8 shows the evaluation result of the pattern in Fig. 4a, plotting only every 25th data point for clarity. A fairly good agreement with the quadratic fit is found.

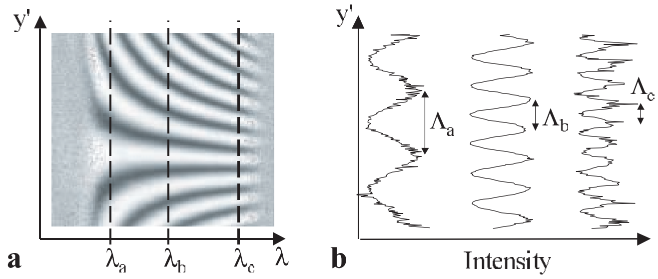

FIGURE 7 Fine evaluation method I. step: Fitting cosine functions to each vertical section of the normalised pattern of Fig. 4a

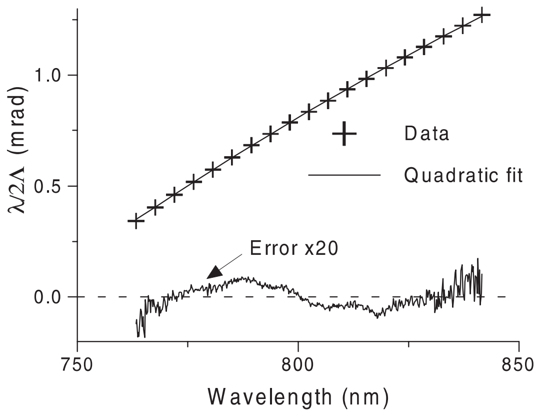

FIGURE 8 Fine evaluation method II. step: Quadratic fit to the the $\lambda/\Lambda(\lambda)$ data of pattern Fig. 4a

The deviation of data points from the quadratic fit is plotted on the same figure, and is magnified by a factor of 20.

The analysis has been carried out on the patterns shown in Fig. 4. First- and second-order angular dispersion coefficients at 800 nm were determined, and are shown in Table 1.

On the accuracy of the measurement we note the following. The noise of the CCD camera has been found to cause a discrepancy less than 0.01 μrad/nm, and the fitting algorithm is accurate to 0.2 μrad/nm. When the measurements have been repeated, the standard deviation was always below 0.2 μrad/nm (see Table 1), confirming that the main limitation is due to the fitting procedure. Accuracy could be further enhanced by using a higher-resolution camera, at the cost of a longer processing time. Experimental data shows that second-order angular dispersion could be determined with the accuracy of 10 nrad/nm^2. We have found that the method is effective for angular dispersion in the range of 0.2–40 μrad/nm, where the upper limit is set by the resolution problem of the high spatial frequency of the fringes.

The accuracy already demonstrated is adequate for controlling or improving the performance of current CPA laser systems. To illustrate this, we add a few examples of optical setups causing angular dispersion of 0.2 μrad/nm at

	Fig. 4a	Fig. 4b	Fig. 4c	Fig. 4d
ε'(μrad/nm)	11.7	11.8	12.0	11.9
ε''(nrad/nm^2)	−32.6	−35.2	−45.2	−41.0

TABLE 1 Values of first- and second-order angular dispersion for the SRI patterns shown in Fig. 4

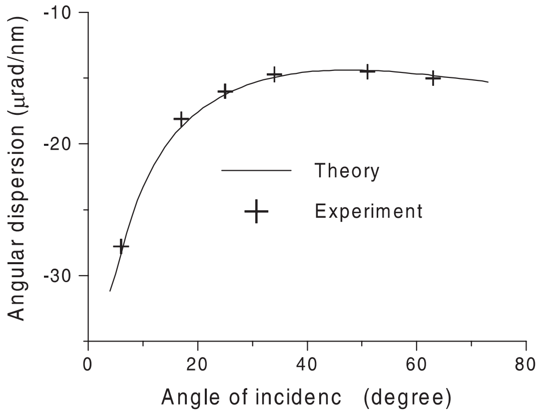

FIGURE 9 Angular dispersion caused by a prism at different angles incidence

the central wavelength of 800 nm. A plane-parallel plate in the laser system (glass filters, Ti:S and nonlinear crystals), with a slight non-parallelism of 13′ for glass SF10, 25′ for sapphire or 39′ for silica, causes this amount of angular dispersion. A pair of identical fused-silica prisms with Brewster angle at minimum deviation in the compressor configuration leaves a residual angular dispersion of 0.2 μrad/nm in the beam if the parallelism alignment is off by 0.257°. Similarly, for a stretcher of $1200\,\mathrm{mm}^{-1}$ gratings in Littrow configuration, the corresponding value is 0.015°.

To test our method, we have measured the angular dispersion of a 45° fused-silica prism as a function of angle of incidence (Fig. 9). The data points are in good agreement with the theoretical calculations, where the Gaussian nature of the incident beam has also been taken into account. As is known [17], the angular dispersion of a Gaussian beam decreases with distance travelled. The lower values of angular dispersion measured in Fig. 4. were achieved using the same prism, but at a different position [18].

5 Conclusions

In this paper we have presented a new method for the measurement of angular dispersion which is based on a spectrally resolved, inverted-side Mach–Zender interferometer. The coarse evaluation of the interferograms is quick, simple and effective for angular dispersion between 10 and 300 μrad/nm, but provides a limited precision only. For the final and most precise alignment of a stretcher-grating system, a more sensitive evaluation method was developed, which has further advantages: it is single-shot and allows for real-time monitoring. Accuracy as high as 0.2 μrad/nm has been achieved for angular dispersion in the range of 0.2–40 μrad/nm.

ACKNOWLEDGEMENTS This work was supported by FKFP 0170/2001 and OTKA #33018.

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